

Phonon-Glass Electron-Crystal (PGEC) Behaviour in Nanowire Based Solid State Heat Pumps for Residential Cooling

Edmund Lew Teng Jun
University of California, Berkeley
Department of Mechanical Engineering
edmundlew@berkeley.edu

As global cooling demands rise, traditional vapor-compression heat pumps face increasing scrutiny due to their reliance on high Global Warming Potential (GWP) refrigerants and mechanical complexity. This paper investigates a sustainable solid-state alternative, which is a nanowire-based thermoelectric heat pump. Nano structuring allows for the decoupling of electrical and thermal transport, achieving an effect known as Phonon-Glass Electron-Glass (PGEC).

The paper analyses the physical mechanisms of the Peltier effect, the role of geometric parameters and surface roughness in suppressing thermal conductivity, and the architectural requirements to build a Pi-couple device for nanowires. Four nanofabrication methods – Metal Assisted Chemical Etching, Reactive Ion Etching, Vapor-Liquid-Solid growth and template assisted electrodeposition are surveyed and evaluated against their cost, precision and material flexibility.

A comparative analysis is made at room temperature of 300K for silicon, bismuth telluride and silicon carbide, with ZT against diameter data drawn from literature. Bismuth telluride nanowires are shown to offer highest peak performance, and silicon carbide nanowires are ruled out due to its suitability for high temperature use cases. The analysis briefly discusses material abundance, supply security and environmental impact, showing how material selection is not just based on efficiency, but rather in terms of deployment viability.

Key engineering challenges to overcome are discussed, including contact resistance, mechanical fragility under thermal cycling and scalability of fabrication from laboratory samples to large scale devices.

The paper concludes that silicon nanowires, despite a lower peak ZT than bismuth telluride, represent the appropriate material for residential-scale solid-state cooling — not because they are the most efficient thermoelectric, but because they are the only candidate whose supply chain can support deployment at the scale required to displace vapor-compression cooling globally.

Keywords—Nanowires, Heating Ventilation & Air Conditioning (HVAC), Phonon-Glass Electron-Crystal (PGEC), Peltier Effect, Figure of Merit (ZT), Solid-state Cooling.

I. INTRODUCTION

The global demand for refrigeration and air conditioning is projected to triple by 2050, creating a critical tension between human comfort and environmental sustainability [1]. The percentage of households with residential AC is projected to

increase from 27% to 41%, with emissions of carbon dioxide between 590 and 1365 million tons of carbon dioxide equivalent ($MtCO_2e$). Current heating, ventilation and air conditioning (HVAC) technologies rely almost exclusively on vapor compression cycles [2]. While such systems are effective, they suffer from two major drawbacks: the use of high Global Warming Potential (GWP) chemical refrigerants like hydrocarbons and hydrofluorocarbons and the mechanical complexity of compressor machinery itself.

A compelling alternative to the status quo is solid state thermoelectric cooling, based on the Peltier effect [3]. Such systems utilize an electric current to physically transport heat across a semiconductor junction. This results in a silent, scalable cooling mechanism that is free of moving parts and harmful gases. However, despite their promise, traditional bulk thermoelectronics have remained a niche technology due to their low efficiency, capped by the fundamental link between electrical thermal conductivity. The efficiency of a thermoelectric material is shown by the figure of merit (ZT) [4] where S is the Seebeck coefficient, σ is the electrical conductivity, T is the absolute temperature and κ is the total thermal conductivity:

$$ZT = \frac{S^2 \sigma T}{\kappa} \quad (1)$$

In bulk semiconductors, σ and κ are proportional to each other, as explained via the Wiedemann-Franz Law [5], which states that an increase in charge carrier concentration must simultaneously enhance thermal transport. This is problematic for solid state thermoelectric heat pumps as pumped heat is rapidly conducted back across the material, significantly reducing its efficiency.

To bypass these fundamental limitations, this paper studies the exploitation of one-dimensional nanowire arrays on a solid-state heat pump. By engineering materials at the nanoscale, it becomes possible to achieve a Phonon-Glass Electron-Crystal (PGEC) state. This approach utilizes boundary scattering which selectively impedes the transport of phonons (thermal vibrations) while maintaining high mobility for charge carriers (electrons). Understanding the design and fabrication of these nanostructured interfaces is critical to realizing the next generation of efficient, solid-state heat pumps.

II. DESIGN

A. The Peltier Effect

The operational core of this system is a Peltier effect that is slightly modified such that an electric current (I) is used to drive a directional heat flux (Q) across **semiconductor junctions** [6]. Unlike metallic conductors where heat is transferred through conduction and vibration of molecules, semiconductors facilitate significant energy transport by utilizing charge carriers as the medium for thermal energy flux.

1) Energy Transitions

At the cold junction, electrons must absorb thermal energy from the lattice to transition from the metal contact into the semiconductor's higher energy conduction band. This is an endothermic process that takes heat away from the metal contact. Conversely at the hot junction, electrons release this potential energy as heat upon returning to the metallic state.

2) Necessity of Dual Carriers (N & P types)

A functional heat pump requires an architecture that utilizes both n-type and p-type materials. If only one of these materials were used, the returning electrical path would just carry heat back to the source, cancelling out the cooling effect.

- **N-Type (Negative):** Characterized by an excess of electrons, which flow opposite to conventional current.
- **P-Type (Positive):** Characterized by an excess of holes, which flow with the conventional current.

3) Pi-Couple Architecture

The different types of materials are arranged in a Pi-couple geometry which means that the device is connected in series electrically but parallel thermally. This allows the electric current to flow in a complete circuit but force the heat to be carried in one direction towards heat sink only.

The total heat transfer can be expressed as [7]:

$$Q = (S_p - S_n)TI \quad (2)$$

where S_p and S_n are the Seebeck coefficients of the p-type and n-type materials, T is the temperature difference between heat source and sink, and I is the current. *Figure 1* below shows the architecture of a thermoelectric heat pump with semiconductor materials [8].

B. The Figure of Merit and PGEC Effect

The performance of a solid-state heat pump is dictated by the dimensionless figure of merit [4] as shown in *Equation 1*. This metric quantifies the efficiency of converting energy into thermal flux while maintaining a temperature gradient.

The issue with using a traditional bulk semiconductor to carry heat between both plates is because σ and κ are proportional to each other from the Wiedemann-Franz Law. Increasing the concentration of charge (σ) leads to a proportionate increase in thermal conductivity (κ). This causes

high thermal-back leakage where heat pumped to the heat sink is rapidly conducted back, neutralizing the cooling effect.

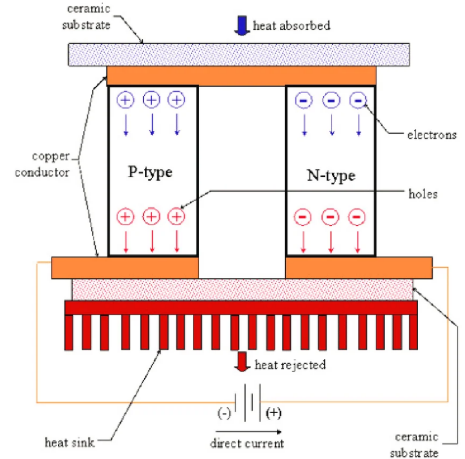


Fig. 1. Architecture of a thermoelectric heat pump with p-type and n-type semiconductor to facilitate heat flow in one direction.

To bypass the limitation, the paper proposes utilizing the PGEC effect, which is a material trait characterized by high electrical mobility of a crystal alongside a low thermal transport coefficient of an amorphous glass. Combining these effects leads to a higher ZT value whereas the material can conduct heat through flow of electricity efficiently without any flowback of heat energy.

Nanowires have been proposed as an ideal material to achieve the PGEC state, as they are able to exploit the disparity of mean free paths between heat-carrying phonons and charge carrying electrons [9]. On the nano scale, heat transfer is reduced significantly due to two major effects of phonon confinement and boundary scattering [10]. The phonons collide with the walls of the nanowire, restricting their travel and limiting the average distance they travel (mean free path). The reduced diameter causes thermal conductivity to drop by a factor of two orders of magnitude, as compared to their bulk values as the high surface area to volume ratio increases frequency of collision. Other than the nanowire diameter, it is suggested that the material roughness and type may also play a role in decreasing the thermal conductivity [11].

At the same time, the smaller wavelength of electrons allows them to transverse through the core of the nanowires unimpeded. This selective scattering decouples the material properties defined by the Wiedemann-Franz Law, allowing for the ZT value to increase significantly beyond its bulk limits.

C. Proposed Design Architecture

To translate the theoretical advantages of the nanowire PGEC effect onto a functional device, the proposed design suggests a modified Pi-couple layout from *Figure 1*. In this configuration, traditional bulk semiconductor blocks are replaced with high-density, vertical nanowire arrays. These nanowires must be made of semiconducting material that can be doped accordingly into p-type and n-type to allow for flow

of electricity and heat in intended directions. This proposed design serves as the bridge between nanoscale transport physics and macroscale thermal management.

The device is constructed as a multi-layer structure with the following components:

- **Thermoelectric Legs:** The core of the device consisting of alternating arrays of n-type and p-type nanowires. The vertical orientation in the design maximises the effective surface area for heat transfer. Depending on the use case, a large density of billions of nanowires per square centimetre may be required to achieve the heat flux necessary for practical cooling applications.
- **Heat Plates:** The nanowires are bonded between two ceramic plates [12], made of materials like alumina (Al_2O_3) or aluminium nitride (AlN). These plates are chosen due to their high thermal conductivity and electrical insulation, which allows heat to flow easily into the nanowire forest while preventing electrical shorts. Copper interconnects bridge the n-type and p-type legs to complete the circuit.
- **Integrated Circuitry:** The design also connects the legs in series electronically and in parallel thermally. This means that the current flows from the positive terminal into the n-type leg and up, across the copper interconnect, down through the p-type leg and back into the negative terminal. The electricity flows in one direction while the heat flows parallel into the heat sink.

This integrated structure results in a solid-state structure that is scalable, simply by increasing size of heat plates or density of the nanowire array.

III. NANOWIRE FABRICATION

To produce the vertical nanowire array described in Section II, high precision nanofabrication techniques are realised, which are then classified into top-down (subtractive) and bottom-up (additive) approaches. The choice of fabrication method directly influences the nanowires' surface morphology, crystal quality and most importantly, its thermoelectric efficiency.

A. Top Down Fabrication Methods

(a) Metal Assisted Chemical Etching (MACE)

MACE is a wet-chemical process in which a single-crystal silicon wafer is selectively etched in regions defined by a noble crystal catalyst. Silver, gold or platinum nanoparticles are first deposited onto the silicon wafer surface, through electroless reduction from a metal-salt solution. The wafer is then immersed in a solution of hydrofluoric acid (HF) and hydrogen peroxide (H_2O_2) where the metal catalyzes the local oxidation and dissolution of silicon beneath each nanoparticle [13].

The metal nanoparticles then sink vertically into the wafer leaving unprotected silicon between them which forms vertical nanowires. The process is relatively cheap, can be run

at room temperature and scales to full sized wafers. However, this process is largely restricted to silicon and produces wires with *stochastic sidewall roughness* because of the etching mechanism [14].

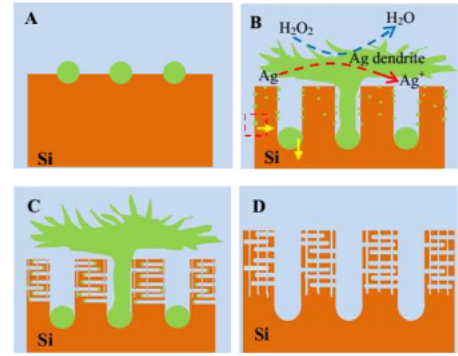


Fig. 2. Schematic illustrating the MACE process with silicon wafer and silver nanoparticles from [13].

(b) Reactive Ion Etching (RIE)

RIE is a dry-plasma process in which a wafer is patterned through a lithography method. This is done by first applying a photoresist hard mask of SiO_2 or a metal, and then exposing the wafer to a reactive plasma which etches the exposed wafer vertically [15]. RIE offers far higher dimensional precision than MACE, because the nanowire diameters and array geometries are defined by the lithography and not the deposition of the catalyst. The trade off for this increased accuracy is its high cost and surface quality. RIE requires specialised lithographic equipment and vacuum systems which scale poorly to larger dimensions. Ion bombardment during the plasma reaction can also cause sub-surface damage that can degrade carrier mobility unless followed by chemical cleaning.

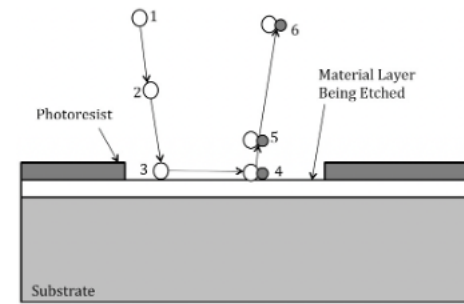


Fig. 3. Schematic illustrating the RIE process where reactive species are adsorped onto the material and react to etch the nanowire [15].

B. Bottom Up Fabrication Methods

(a) Vapor-Liquid-Solid (VLS) Growth

VLS is the go-to fabrication method for semiconductor nanowires, proposed by Wagner and Ellis in 1964 to grow defect free structures [16]. Metal nanoparticles like gold are deposited onto a substrate and heated above the metal-semiconductor eutectic temperature to form liquid alloy droplets. A gaseous precursor like silane (SiH_4) is introduced via chemical vapor deposition (CVD), which then decomposes

at the droplet surface and dissolves into it until it is supersaturated. At this point, the solid semiconductor precipitates at the droplet-substrate interface and grows vertically upwards as a nanowire. The wire diameter is determined by catalyst droplet size. VLS produces smooth, single-crystal wires with excellent dimensional control, but is constrained by the high process temperature and the risk of catalyst contamination.

(b) Template Assisted Electrodeposition (AAO Templates)

For materials that cannot be grown by VLS, especially tellurium-based compounds like Be_2Te_3 , template assisted electrodeposition provides an alternative method. An anodic aluminum oxide membrane (AAO) containing a high-density array of cylindrical pores is used as a sacrificial scaffold.

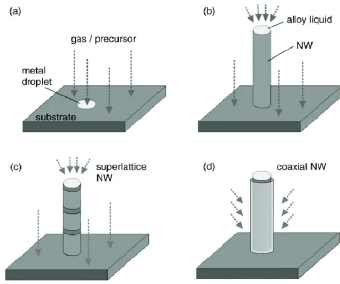


Fig. 4. Schematic illustrating the VLS process where the liquid alloy droplet is superstuated by the precursor and precipitates vertically to form the nanowire [16].

Material is electrodeposited from solution into the pores, filling them up from a conductive backing layer. The template is then dissolved chemically to form a dense vertical nanowire array [18]. Template assisted methods offer the highest array densities of all techniques and can accommodate compound semiconductors which resist gas-phase synthesis. However, they produce polycrystalline wires rather than single-crystal wires which limits carrier mobility.

C. Summary

Each of these four methods discuss the trade-off between cost vs precision and material flexibility vs crystallinity. MACE and electrodeposition are cost-effective and scalable, but they produce wires with defects. RIE and VLS produce high-quality wires but are restricted by material constraints. The appropriate manufacturing method depends on the ideal nanowire material identified in the following section.

IV. ANALYSIS: COMPARATIVE EVALUATION OF NANOWIRE MATERIALS

The thermoelectric performance of a nanowire is largely determined by the intrinsic transport properties of its constituent semiconductor material. While the PGEC framework in Section II applies to any nanostructure semiconductor, the ZT value at room temperature can vary with different materials and diameters. This section examines the three materials of interest, which are silicon (Si), bismuth

telluride (Bi_2Te_3) and silicon carbide (SiC), and evaluates their viability for room temperature operation at 300K.

(a) Silicon Nanowires

Bulk silicon is a poor thermoelectric material, with $ZT = 0.01$ at room temperature [19] due to its high lattice conductivity of roughly $150 W/m \cdot K$. However, two landmark studies from [19] and [20] show that downsizing bulk silicon to the nanoscale fundamentally changes its thermal behavior.

In [19], Hochbaum et.al reports that rough silicon nanowires produced through aqueous electroless etching achieves $ZT \approx 0.6$ at 300K for 50nm diameter nanowires, due to suppression of lattice thermal conductivity through phonon boundary scattering. In [20], Boukai et al. demonstrates smooth, VLS grown silicon nanowires with rectangular cross sections of $10 \times 20 nm$ and $20 \times 20 nm$ can achieve $ZT = 0.36$ and 0.22 respectively at 300K, with peak performance at 200K for larger wires. Converting the rectangular wires to an effective diameter with Equation (2), a graph of comparisons for Si nanowires is shown in the figure below:

$$d_{eff} = \sqrt{\frac{4A}{\pi}} \quad (2)$$

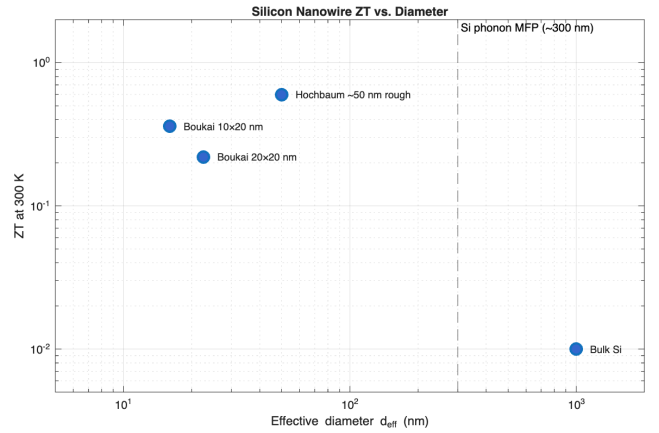


Fig. 5. Graph plotting the ZT value at 300K against the effective diameter of silicon nanowires obtained from literature research.

Figure 5 shows that the ZT value at 300K does not scale directly with diameter alone. Hochbaum's 50 nm rough nanowires outperforms Boukai's smaller and smoother wires, indicating that *surface morphology plays a crucial role to PGEC behaviour*. By using electroless etching to produce rough sidewalls, phonons across a broad spectrum of wavelengths are scattered, suppressing thermal conductivity to its amorphous limit while allowing electron transport [21]. This favours a MACE-based synthesis, where roughness emerges naturally from the etching mechanism, over smoother bottom-up routes such as VLS.

Below diameters of 7 nm , atomistic tight-binding calculations suggest that quantum confinement may further enhance the power factor at room temperature, but these gains are offset by a degradation of electron mobility [22]. For useful analysis, the practical operation window is the 30 nm to 100 nm range which is accessible via MACE, where rough silicon nanowires combine near-bulk electrical conductivity with amorphous limit thermal transport.

(b) Bismuth Telluride Nanowires

For bulk materials, Bi_2Te_3 is the benchmark room-temperature thermoelectric material, achieving a ZT value of 1. By confining it to the nanometer scale, its performance becomes substantially better.

According to research from [23], theoretical analysis using a constant-relaxation-time approximation predicts that p-type (110)-oriented Bi_2Te_3 nanowires achieve room-temperature ZT values of 1.2, 1.3 and 1.7 for square cross-sections of $30 \times 30\text{ nm}$, $15 \times 15\text{ nm}$ and $7 \times 7\text{ nm}$ respectively. The main driving factor for this is due to carrier confinement effects that boost the Seebeck coefficient as the cross section drops below 30 nm . Experimentally, Menke et al. [24] synthesized Be_2Te_3 nanowire arrays with cyclic electrodeposition into anodic aluminium oxide (AAO) templates, with diameters between 30 and 150 nm , achieving ZT values from 0 to 0.85.

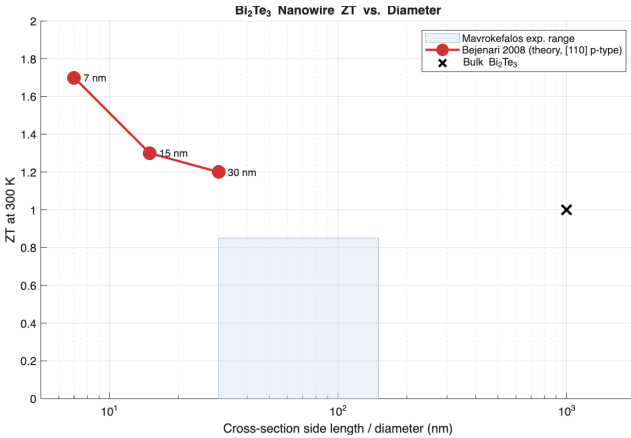


Fig. 6. Graph plotting the ZT value at 300K against the diameter of bismuth telluride obtained from literature research.

The gap between theoretical prediction and experimental measurement is consistent with literature of nanostructure thermoelectronics. It is shown that ZT values typically fall to 60% to 70% of theoretical values due to defects, contact resistance and surface oxidation. Even with this gap, Bi_2Te_3 nanowires offer a clear performance advantage over silicon at room temperature of 300 K . However, the trade-offs in material availability and fabrication complexity that come with this improvement are addressed in Section V.

The figure below shows the comparison of ZT values across different literatures for Bi_2Te_3 nanowires:

(c) Silicon Carbide Nanowires

Silicon carbide is frequently cited as a sustainable, mechanically robust alternative for conventional thermoelectric semiconductor, hence its inclusion in this comparison. However, evidence from research shows that SiC nanowires are unsuitable for room-temperature cooling applications. Experimental measurements on single $\beta - \text{SiC}$ nanowires show ZT values on the order of 10^{-3} to 10^{-4} at 300 K , which is several orders of magnitude below the threshold required for any practical heat pump [25]. Even in theoretical studies assuming idealized $2 \times 2\text{ nm}$ cross sections predict a maximum ZT of $0.65 - 0.89$, and this is only achieved at temperatures of 500 K [26], making it irrelevant for our use case.

The underlying issue has to do with the inherent properties of SiC . It has a wide bandgap of 3 eV and a high optical phonon energies which suppress heat transport at room temperature. Its high bulk thermal conductivity also means that even with aggressive nanostructuring, lattice conductivity remains dominant. SiC may be useful for heat recovery at temperatures above 800 K , but for room temperature cooling, they are not as effective as the two prior materials.

V. SUSTAINABILITY AND AVAILABILITY

Material analysis in Section IV establishes that Bi_2Te_3 nanowires offer better performance at room temperature cooling, with theoretical ZT values that are two to three times higher at the same diameter range. However, the choice of thermoelectric material cannot be made based on efficiency alone. To displace vapor compression cooling globally, constraints on material abundance, supply security and environmental impact have to be considered in the material selection.

A. Abundance and Supply Risk

There exists a huge disparity between the supply of silicon and tellurium. Silicon is the second most abundant element in the Earth's crust at approximately 28% by mass [27], while tellurium is one of the rarest stable elements with a crustal abundance of about 3 parts per billion [28]. This represents a difference of approximately eight orders of magnitude.

At the same time, tellurium is produced almost exclusively as a byproduct of copper refining, meaning that its supply is mainly dictated by the demand for a completely unrelated commodity. The European Union and United States both classify tellurium as a critical raw material with high supply risk, highlighting its scarcity and geographical concentration of production [29].

A residential heat pump system targeting global deployment would require billions of nanowires per square centimeter across square meter devices. At this scale, it can be easily expected that the supply will not be able to satisfy the demand, hindering the technology's progress to widescale implementation.

B. Vapor-Compression Cycle

In terms of sustainability, the technology that we are replacing sets a low bar. Conventional residential air

conditioning uses R-410A, a hydrofluorocarbon (HFC) blend with a 100-year global warming potential (GWP) of 2088 [30]. Relevant US policy aims to phase down R-420A by 85% between 2022 and 2036 with newer alternatives such as R-32 (GWP 675) or R-454B (GWP 466) as transitional elements.

The technology that this paper is proposing, solid-state thermoelectric heat pumps eliminate the need for refrigerant entirely, relying only on doped semiconductor and metallic contacts as their working materials. It must be noted that the embodied life-cycle impact of nanowire fabrication from wafer production, chemical etching in device assembly and end-of-life recovery must still be accounted for in a complete comparison. However, the absence of a high-GWP working fluid leaves solid-state systems with a structural advantage that is surely more environmentally friendly compared with vapor-compression cycles.

Having now considered both A and B, this paper is able to offer a more comprehensive review on material selection. Silicon nanowires can offer ZT of 0.6 at room temperature, which is sufficient to deliver meaningful cooling without the environmental impacts brought by conventional HVAC systems. The silicon material is also essentially unconstrained in supply and supported by mature wafer-scale fabrication infrastructure. Bi_2Te_3 nanowires offer higher peak performance, but at a material supply cost that is incompatible with deployment at a global scale. For the application of focus in this paper, silicon is therefore the appropriate choice not because it offers the best performance, but because it is the only sustainable candidate whose supply chain can support the volumes required.

VI. CHALLENGES

Through analysis in Sections IV and V, it is established that rough silicon nanowires manufactured through MACE is the ideal material for the proposed solid state residential cooling system. However, several challenges remain between the current analysis and a working device. These challenges are not fundamental physics problems, as the PGEC mechanism itself is well established, but rather practical issues to do with fabrication, durability and scaling that would have to be addressed for the nanowire heat pumps to replace conventional vapor compression systems.

A. Contact Resistance

The largest source of inefficiency in practical nanowire thermoelectric devices is the resistance at the interface between macroscopic copper electrodes and individual nanowires. A device with billions of wires per square centimeter contains a large number of metal-semiconductor junctions, each of which dissipates a small amount of energy as ohmic heating. This is the reason for the 30% to 40% difference between theoretical and experimental ZT values in literature from [23] and [24].

To ensure that a practical device is as efficient as possible, careful engineering of the metallization process must be carried out, through silicide formation or other low-barrier

interface treatments. This is an active area of research without a clear consumer-scale solution, that if solved would benefit the proposed device architecture.

B. Mechanical Fragility

A vertical nanowire forest is mechanically delicate due to its high aspect ratio (ratio between diameter and length). The wires must first survive the assembly process, which includes bonding to ceramic plates and application of copper interconnects. It then has to survive through at least a decade of operational stress which includes thermal expansion, vibration and mechanical handling. Shear stresses may be introduced when the copper interconnects, ceramic heat plates and silicon nanowires expand differently under thermal stress. This can cause delamination or fracture over repeated thermal cycles. Most work on nanowires currently are at the laboratory scale under highly controlled conditions, hence the long-term reliability of nanowire arrays under HVAC duty cycles remains largely unverified.

C. Scalability

Translating results from a square-centimetre research sample into a square-meter HVAC device requires precise control over doping uniformity, surface morphology and defect density across an area that is six orders of magnitude larger than current demonstrations. In principle, MACE is wafer scalable, but maintaining uniform sidewall roughness and consistent diameters over a 300 mm wafer is challenging. Cost-effective manufacturing would require technology to be developed with existing semiconductor fabrication infrastructure, to speed up the route to industrial deployment.

VII. CONCLUSION

This paper has examined the viability of nanowire-based solid state heat pumps as a sustainable alternative to vapor-compression cooling for residential applications. The central insight is that nano structuring breaks the Wiedemann-Franz coupling between electrical and thermal conductivity, producing a PGEC material. This material exhibits the electrical conductivity of a crystal but the thermal conductivity of an amorphous glass simultaneously. Nanowires are therefore the solution that can break the barrier constraining bulk semiconductors in solid state cooling systems.

A comparative analysis of three nanowire materials at room temperature of 300K shows that bismuth telluride offers the highest ZT value, silicon offers strong but lower peak performance and silicon carbide is unsuitable for residential operating temperatures. Some critical relationships discovered is that ZT in silicon nanowires does not scale linearly with a reduction in nanowire diameter. However, surface morphology can play a crucial role, with rough sidewalls produced through electroless etching outperforming smaller but smooth nanowires. This observation therefore favours Metal-Assisted Chemical Etching (MACE) as the best fabrication route.

When material selection criteria are extended to include material abundance, supply security and environmental impact, the choice of material becomes clearer. Silicon is 28% of the Earth's crust, and benefits from already mature wafer-scale fabrication infrastructure. Tellurium at 3 parts per billion of the Earth's crust is incompatible with the volumes required to displace residential cooling systems. The conclusion this study comes to is therefore that silicon is the best candidate for the proposed device despite it slightly lacking in the ZT value.

Significant challenges remain in the form of contact resistance, mechanical durability and manufacturing scalability, but these are integration problems rather than barriers in the underlying physics itself. The proposed device, consisting of MACE-fabricated silicon nanowire arrays bonded between ceramic heat plates in a Pi-couple configuration, represents a refrigerant-free cooling system that can meet the demand for global cooling. Future work should focus on overcoming the contact resistance, mechanical durability testing under realistic HVAC duty cycles and pilot scale fabrication runs to assess uniformity across devices when developed at large sizes.

ACKNOWLEDGMENT

The author would like to thank Professor Liwei Lin for his guidance and instruction provided in MECENG118 at the University of California, Berkeley which formed the foundation for this work.

REFERENCES

- [1] Giacomo Falchetta, Enrica De Cian, Filippo Pavanetto, and I. S. Wing, "Inequalities in global residential cooling energy use to 2050," *Nature Communications*, vol. 15, no. 1, Sep. 2024, doi: <https://doi.org/10.1038/s41467-024-52028-8>.
- [2] F. Alsouda, N. S. Bennett, S. C. Saha, F. Salehi, and M. S. Islam, "Vapor Compression Cycle: A State-of-the-Art Review on Cycle Improvements, Water and Other Natural Refrigerants," *Clean Technologies*, vol. 5, no. 2, pp. 584–608, Jun. 2023, doi: <https://doi.org/10.3390/cleantechnol5020030>.
- [3] D. de Azcuénaga, "Peltier effect: Fundamentals and applications," *PCIM News Platform*, Oct. 14, 2025. <https://news.pcim.mesago.com/peltier-effect-fundamentals-and-applications-a-cf34f8d5a26cb5f4c4b45666f97c2443/>
- [4] R. Jaafreh, K. Yoo Seong, J.-G. Kim, and K. Hamad, "A deep learning perspective into the figure-of-merit of thermoelectric materials," *Materials Letters*, vol. 319, p. 132299, Jul. 2022, doi: <https://doi.org/10.1016/j.matlet.2022.132299>.
- [5] V. Devanathan, "The Wiedemann-Franz Law for Electrical and Thermal Conduction in Metals," vol. 4, pp. 1–26, Jan. 2021, Available: https://www.researchgate.net/publication/355196811_The_Wiedemann-Franz_Law_for_Electrical_and_Thermal_Conduction_in_Metals
- [6] YU. G. GUREVICH and J. E. VELAZQUEZ-PEREZ, "Peltier Effect in Semiconductors," *Wiley Encyclopedia of Electrical and Electronics Engineering*, pp. 1–21, Apr. 2014, doi: <https://doi.org/10.1002/047134608x.w8206>.
- [7] I. Terasaki, "Thermal Conductivity and Thermoelectric Power of Semiconductors," *Reference Module in Materials Science and Materials Engineering*, 2016, doi: <https://doi.org/10.1016/b978-0-12-803581-8.00071-2>.
- [8] Adeyanju Anthony Ademola, "Theoretical and Experimental Analysis of a Thermoelectric Air-conditioning System," *IntechOpen eBooks*, Jun. 2020, doi: <https://doi.org/10.5772/intechopen.88664>.
- [9] R. Chen, J.-H. Lee, W. Lee, and D. Li, "Thermoelectrics of Nanowires," vol. 119, no. 15, pp. 9260–9302, Mar. 2019, doi: <https://doi.org/10.1021/acs.chemrev.8b00627>.
- [10] K. V. Tretiakov and K. Hyžorek, "Role of the phonon confinement effect and boundary scattering in reducing the thermal conductivity of argon nanowire," *The Journal of Chemical Physics*, vol. 154, no. 5, Feb. 2021, doi: <https://doi.org/10.1063/5.0036572>.
- [11] A. Malhotra and M. Maldovan, "Impact of Phonon Surface Scattering on Thermal Energy Distribution of Si and SiGe Nanowires," *Scientific Reports*, vol. 6, no. 1, May 2016, doi: <https://doi.org/10.1038/srep25818>.
- [12] M. K. Shilpa, M. Abdul Raheman, A. Aabid, M. Baig, R. K. Veerasha, and N. Kudva, "A Systematic Review of Thermoelectric Peltier Devices: Applications and Limitations," *Fluid Dynamics & Materials Processing*, vol. 19, no. 1, pp. 187–206, 2023, doi: <https://doi.org/10.32604/fdmp.2022.020351>.
- [13] S. Li *et al.*, "Fabrication of porous silicon nanowires by MACE method in HF/H₂O₂/AgNO₃ system at room temperature," *Nanoscale research letters (Online)*, vol. 9, no. 1, Apr. 2014, doi: <https://doi.org/10.1186/1556-276x-9-196>.
- [14] A. A. Leonardi, M. J. L. Faro, and A. Irrera, "Silicon Nanowires Synthesis by Metal-Assisted Chemical Etching: A Review," *Nanomaterials*, vol. 11, no. 2, p. 383, Feb. 2021, doi: <https://doi.org/10.3390/nano11020383>.
- [15] M. Huff, "Recent Advances in Reactive Ion Etching and Applications of High-Aspect-Ratio Microfabrication," *Micromachines*, vol. 12, no. 8, p. 991, Aug. 2021, doi: <https://doi.org/10.3390/mi12080991>.
- [16] R. S. Wagner and W. C. Ellis, "VAPOR-LIQUID-SOLID MECHANISM OF SINGLE CRYSTAL GROWTH," *Applied Physics Letters*, vol. 4, no. 5, pp. 89–90, Mar. 1964, doi: <https://doi.org/10.1063/1.1753975>.
- [17] T. S. Y. Moh *et al.*, "Silicon nanowire FET arrays for real time detection of chemical activation of cells," *2012 IEEE 25th International Conference on Micro Electro Mechanical Systems (MEMS)*, pp. 1344–1347, Jan. 2012, doi: <https://doi.org/10.1109/memsys.2012.6170424>.
- [18] J.-B. Shi, C.-J. Chen, Y.-T. Lin, W.-C. Hsu, Y.-C. Chen, and P.-F. Wu, "Anodic Aluminum Oxide Membrane-Assisted Fabrication of β -In₂S₃ Nanowires," *Nanoscale Research Letters*, vol. 4, no. 9, Jun. 2009, doi: <https://doi.org/10.1007/s11671-009-9357-z>.
- [19] A. I. Hochbaum *et al.*, "Enhanced thermoelectric performance of rough silicon nanowires," *Nature*, vol. 451, no. 7175, pp. 163–167, Jan. 2008, doi: <https://doi.org/10.1038/nature06381>.
- [20] A. I. Boukai, Y. Bunimovich, J. Tahir-Kheli, J.-K. Yu, W. A. Goddard III, and J. R. Heath, "Silicon nanowires as efficient thermoelectric materials," *Nature*, vol. 451, no. 7175, pp. 168–171, Jan. 2008, doi: <https://doi.org/10.1038/nature06458>.
- [21] G. Pennelli, E. Dimaggio, and A. Masci, "Silicon Nanowires: A Breakthrough for Thermoelectric Applications," *Materials*, vol. 14, no. 18, p. 5305, Sep. 2021, doi: <https://doi.org/10.3390/ma14185305>.
- [22] N. Neophytou, M. Wagner, H. Kosina, and S. Selberherr, "Analysis of Thermoelectric Properties of Scaled Silicon Nanowires Using an Atomistic Tight-Binding Model," *Journal of Electronic Materials*, vol. 39, no. 9, pp. 1902–1908, Jan. 2010, doi: <https://doi.org/10.1007/s11664-009-1035-5>.
- [23] Igor Bejenari and B. Г. Канцер, "Thermoelectric properties of bismuth telluride nanowires in the constant relaxation-time approximation," *Physical Review B*, vol. 78, no. 11, Sep. 2008, doi: <https://doi.org/10.1103/physrevb.78.115322>.
- [24] E. J. Menke, M. A. Brown, Q. Li, J. C. Hemminger, and R. M. Penner, "Bismuth Telluride (Bi₂Te₃) Nanowires: Synthesis by Cyclic Electrodeposition/Stripping, Thinning by Electrooxidation, and Electrical Power Generation," *Langmuir*, vol. 22, no. 25, pp. 10564–10574, Aug. 2006, doi: <https://doi.org/10.1021/la061275g>.
- [25] Y. Xing *et al.*, "Principles and Methods for Improving the Thermoelectric Performance of SiC: A Potential High-Temperature Thermoelectric Material," *Materials*, vol. 17, no. 15, pp. 3636–3636, Jul. 2024, doi: <https://doi.org/10.3390/ma17153636>.
- [26] J. H. Choi, M. Pala, L. Latu-Romain, and E. Bano, "Theoretical Study of Thermoelectric Properties of SiC Nanowires," *Materials Science Forum*, vol. 717–720, pp. 561–564, May 2012, doi: <https://doi.org/10.4028/www.scientific.net/msf.717-720.561>.

- [27] Royal Society of Chemistry, "Silicon - Element information, properties and uses | Periodic Table," *Rsc.org*, 2017. <https://periodic-table.rsc.org/element/14/silicon>
- [28] P. Nuss, "Losses and environmental aspects of a byproduct metal: tellurium," *Environmental Chemistry*, vol. 16, no. 4, p. 243, 2019, doi: <https://doi.org/10.1071/en18282>.
- [29] U. Jamil and N. M. Holden, "Assessment of Scarcity, Toxicity, and Circularity Risks in the European Thermoelectric Market: A Focus on Tellurium, Antimony, Bismuth, and Lead," *Clean Technologies*, vol. 7, no. 1, pp. 5–5, Jan. 2025, doi: <https://doi.org/10.3390/cleantechnol7010005>.
- [30] "R-410A," *Wikipedia*, Jan. 12, 2022. <https://en.wikipedia.org/wiki/R-410A>